



The Java Blueprint

A Visual Guide to Variables, Data Types, Math, and Input

Master the foundational building blocks of Java programming.



Memory: The shelf itself.

Variables: Temporary, labeled storage bins.

Data Types: The physical shape of the bin. (You cannot put a word in a bin designed for whole numbers).

```
int age = 18;
```

Variables. Values can change as the program runs.



```
[ score = 10;  
  score = 20; ]
```

Constants (final). Locked upon creation. Prevents accidental changes (like the value of Pi). Named in **ALL_CAPS** by convention.



```
[ final double PI = 3.14159; ]
```



Whole Numbers

byte ^{30%}
short
long
int



Decimals

303 ^{30%}
float ^{30%}
double



Single Characters

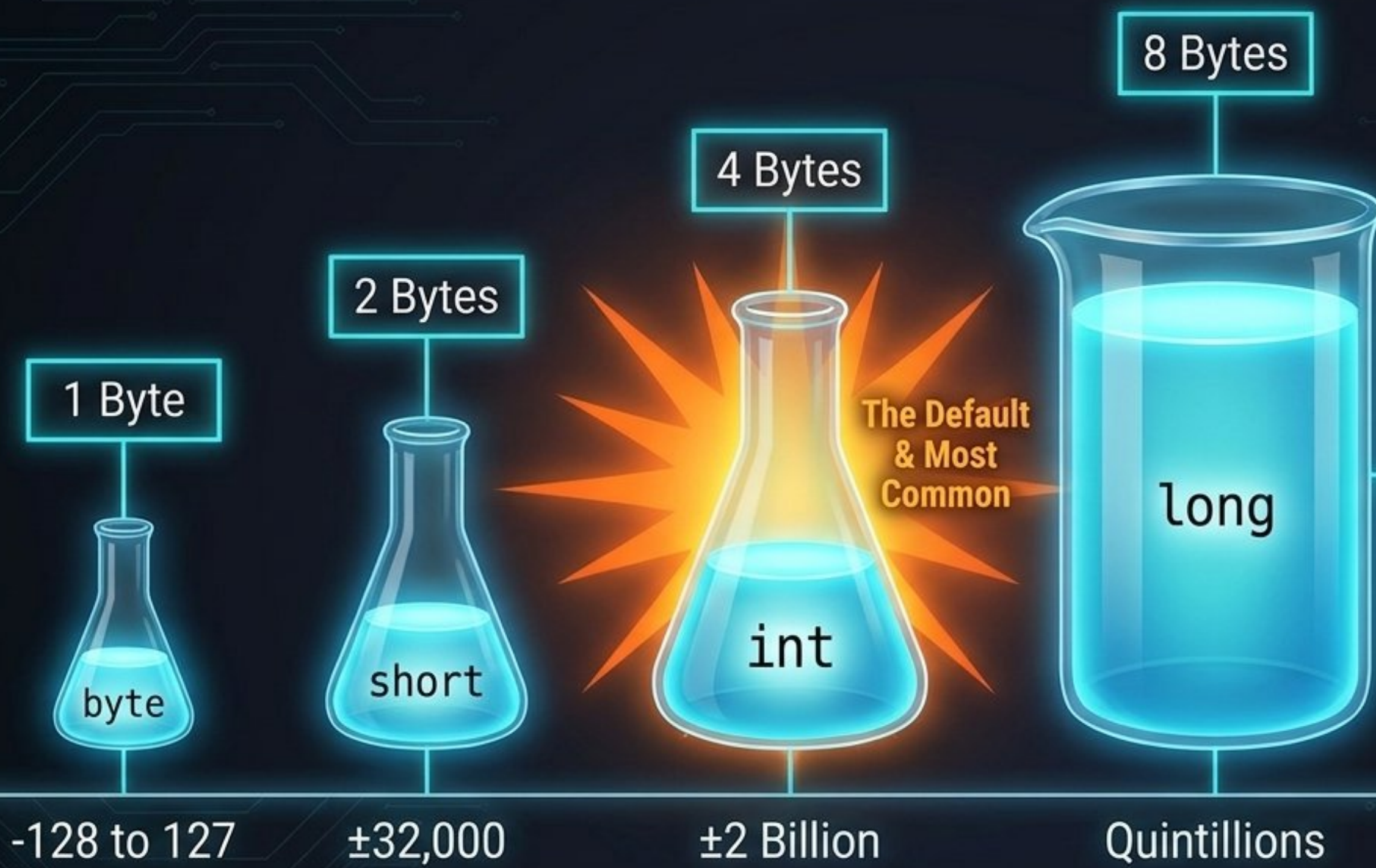
char



Truth Values

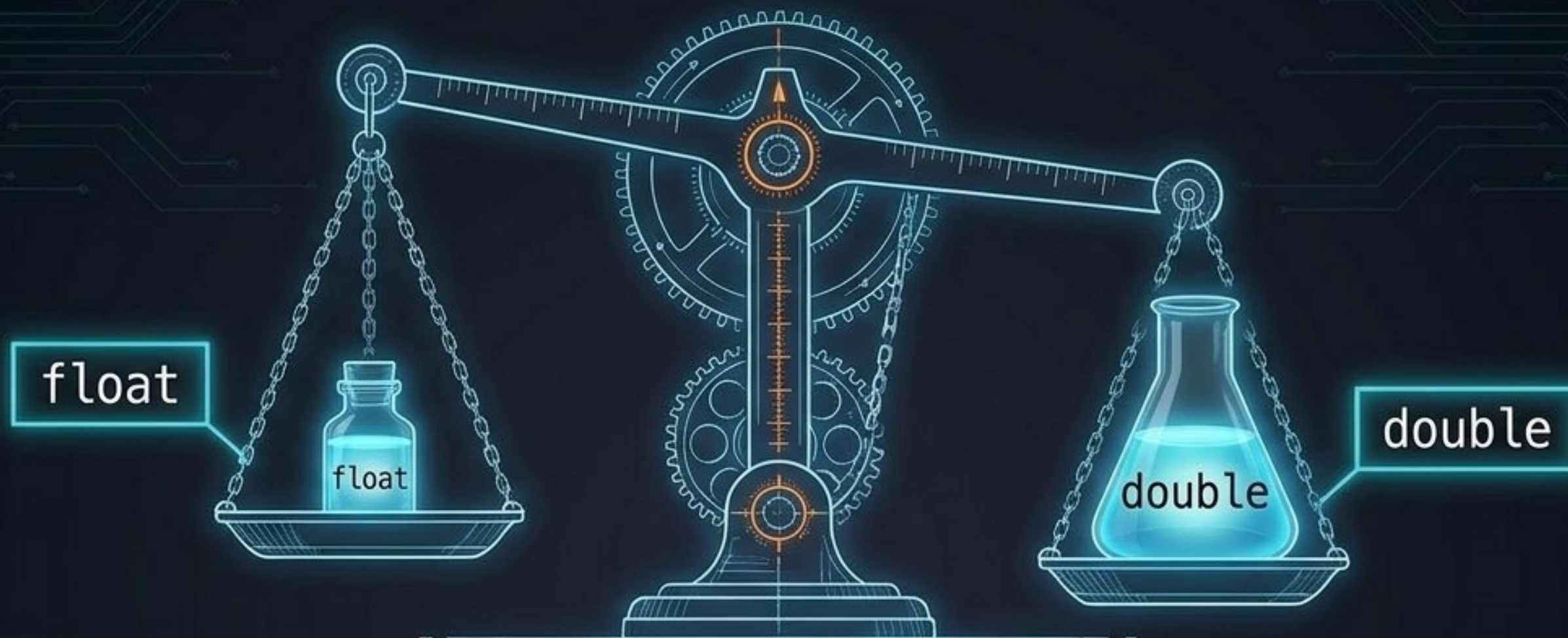
boolean

Primitives are the simplest, fundamental data types in Java.
They store raw values directly in memory.



long needs an 'L' suffix:
`long stars = 9_000_000_000L;`

Pro-Tip: Use underscores to make large numbers readable:
`int budget = 1_000_000;`



float

- 4 bytes (Single precision: ~7 decimal places).
- Rule: Must end with an 'f' or 'F'.

```
float price = 19.99f;
```

double

- 8 bytes (Double precision: ~15 decimal places).
- Rule: The default for decimals in Java.

```
double pi = 3.14159;
```

Takeaway: Use `double` for math requiring high accuracy, but `float` saves memory.



Header: `char`.

Body: Stores exactly one character.
Always uses single quotes.

Code: `char grade = 'A';` or
`char symbol = '@';`



Header: `boolean`.

Body: The foundation of computer logic.
Can only hold two values: `true` or `false`.

Code: `boolean isStudent = true;`

Arithmetic Operators



+ (Addition): $10 + 2 = 12$



- (Subtraction): $10 - 2 = 8$



* (Multiplication): $10 * 2 = 20$



/ (Division): $10 / 2 = 5$



WARNING: Integer Division
Drops Decimals!

$10 / 3$ results in 3 (Not 3.33).

To retain decimals, at least one number
must be a double:

$10.0 / 3$

The Magic of Modulus (%)

Definition: Returns the remainder of division.

$$10 \% 3 = 1$$



Why do we care? It is the ultimate tool for finding even or odd numbers.
If $\text{number} \% 2 == 0$, the number is perfectly even.

Type Conversion (Implicit Widening)



Process: Happens automatically. Java safely upgrades a smaller data type to a larger one because there is plenty of room in memory. No data is lost.

`int whole = 5;` → `double decimal = whole;` → Result: 5.0

Type Conversion (Explicit Narrowing)

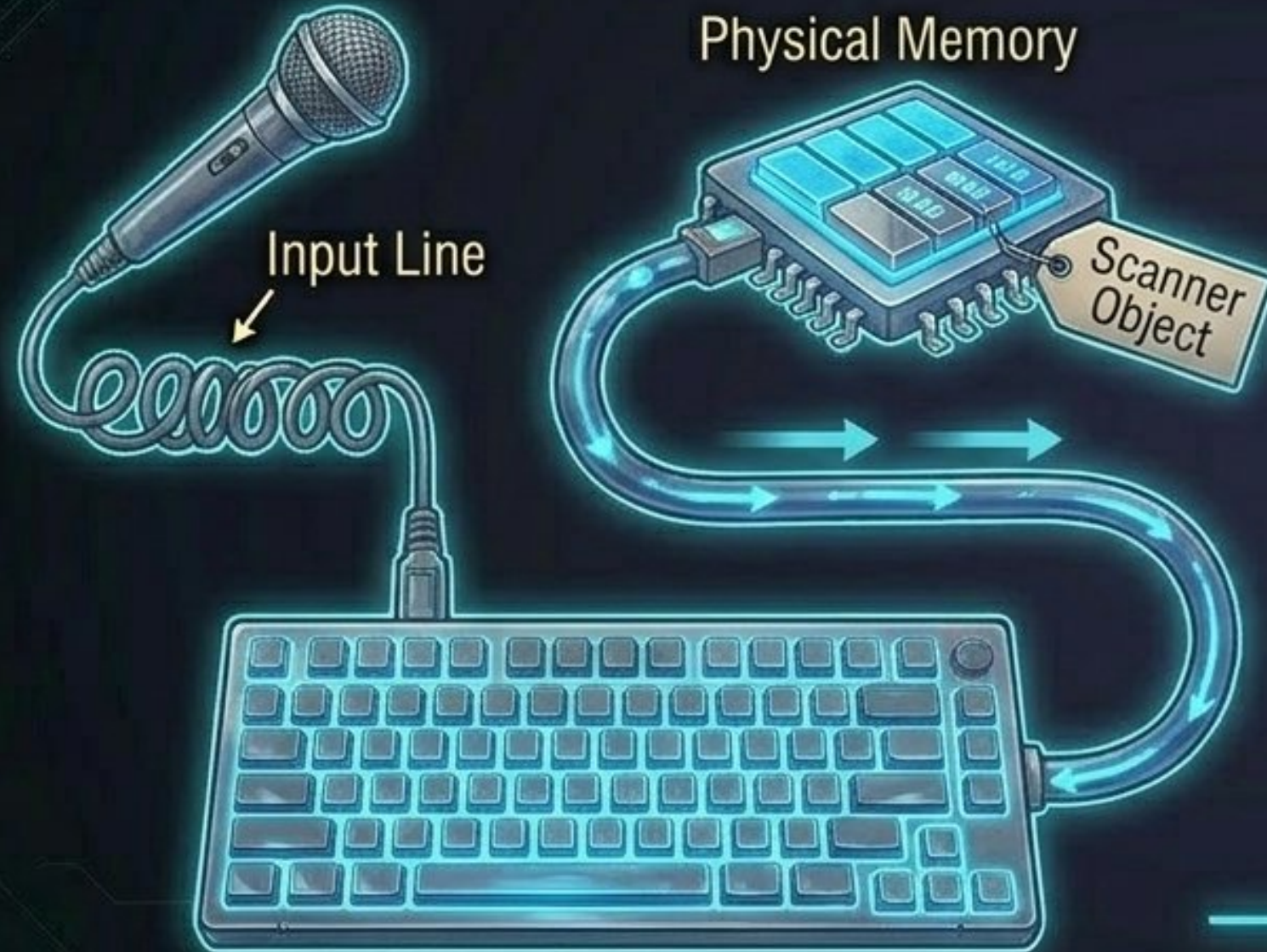
double (8.99)



Process: You must force Java to do this, acknowledging data will be lost. Decimals are chopped off, not rounded!

`double exact = 8.99;` → `int rough = (int) exact;` → Result: 8

Scanner Class (Java Input)



1. Import (Get the blueprint)

```
import java.util.Scanner;
```



2. Create (Plug it in)

```
Scanner input = new Scanner(System.in);
```



3. Listen (Capture the data)

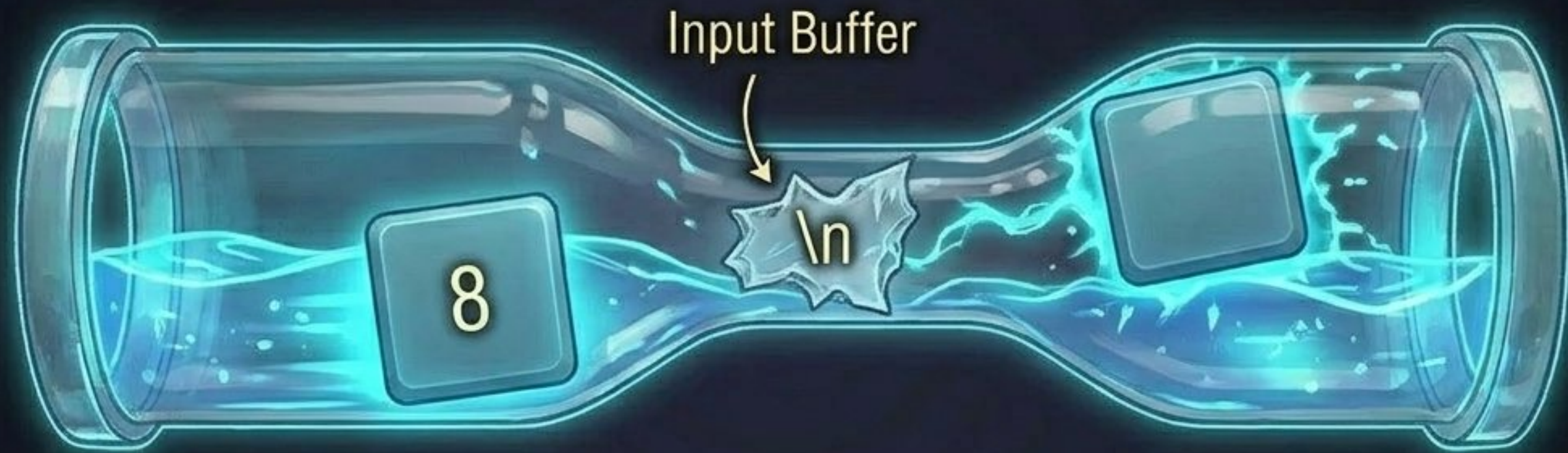
```
int age = input.nextInt();
```

or

```
String name = input.nextLine();
```

Pro-Tip: Always close your scanner when finished (`input.close();`) to free up system resources.

The Scanner Gotcha: Clearing the Buffer



(The Bug): Using `nextInt()` leaves the 'Enter' keystroke in the buffer. If you follow it with `nextLine()`, the program sees that 'Enter', thinks you finished, and skips your prompt!

(The Fix): Add a blank `input.nextLine();` to act as a 'plunger' and flush the buffer before asking for the next string.

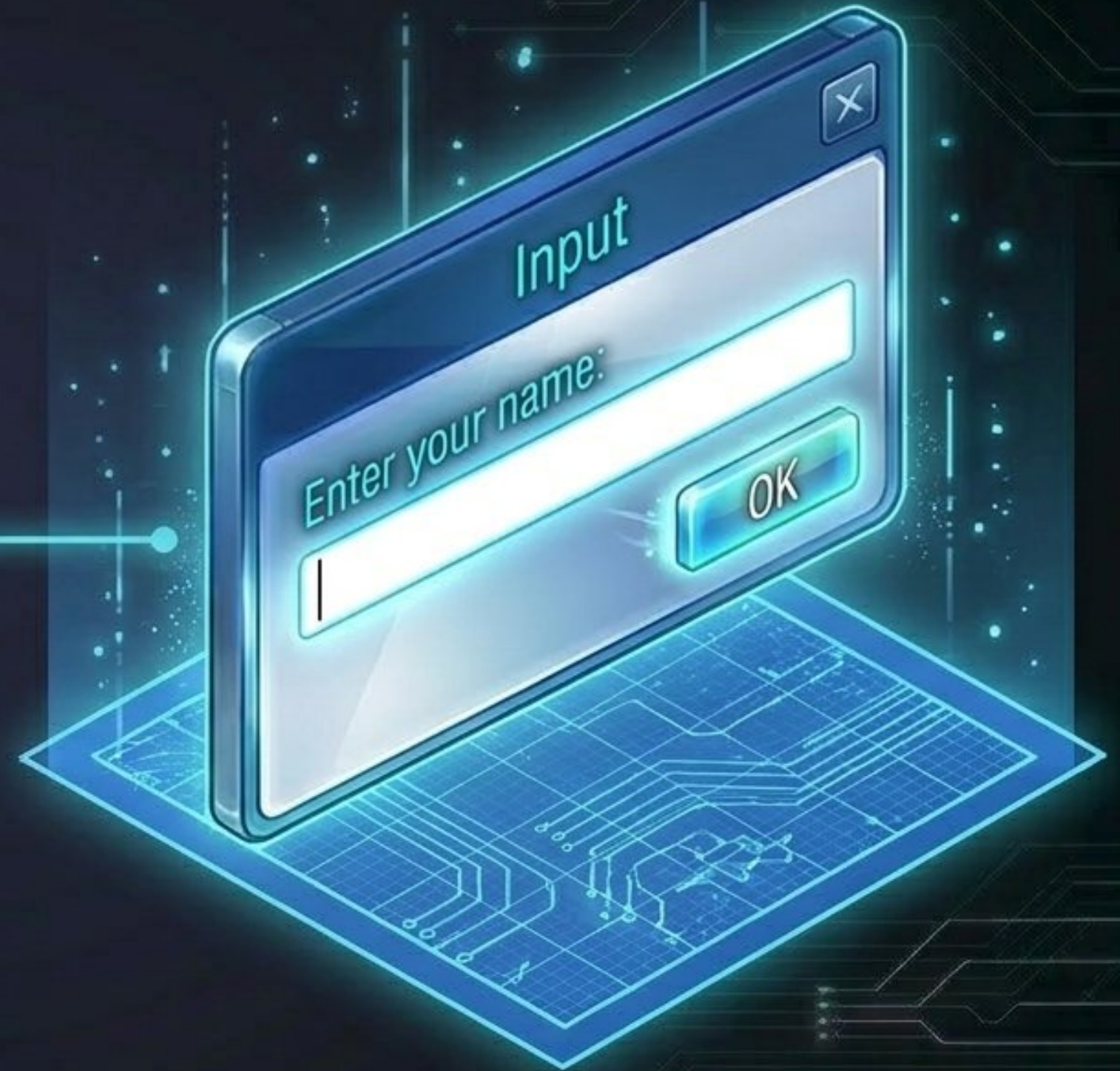
Graphical Input – JOptionPane

Want to escape the text console? Use JOptionPane for Graphical User Interfaces (GUI).

```
import javax.swing.JOptionPane;  
  
String name = JOptionPane.  
    showInputDialog("Enter your name:");
```



Warning: GUI inputs always return Strings.
To get a number, you must parse it:
`int age = Integer.parseInt(input);`



Synthesis

The
Communication
Line

```
import java.util.Scanner;

public class CircleArea {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        final double PI = 3.14159;

        System.out.print("Enter radius: ");
        double radius = input.nextDouble();

        double area = PI * (radius * radius);
        System.out.println("Area: " + area);
        input.close();
    }
}
```

The Locked
Constant

The Decimal
Material

The Toolbelt
(Math)

You now have the containers, the materials, the tools, and the communication lines. It's time to start building.